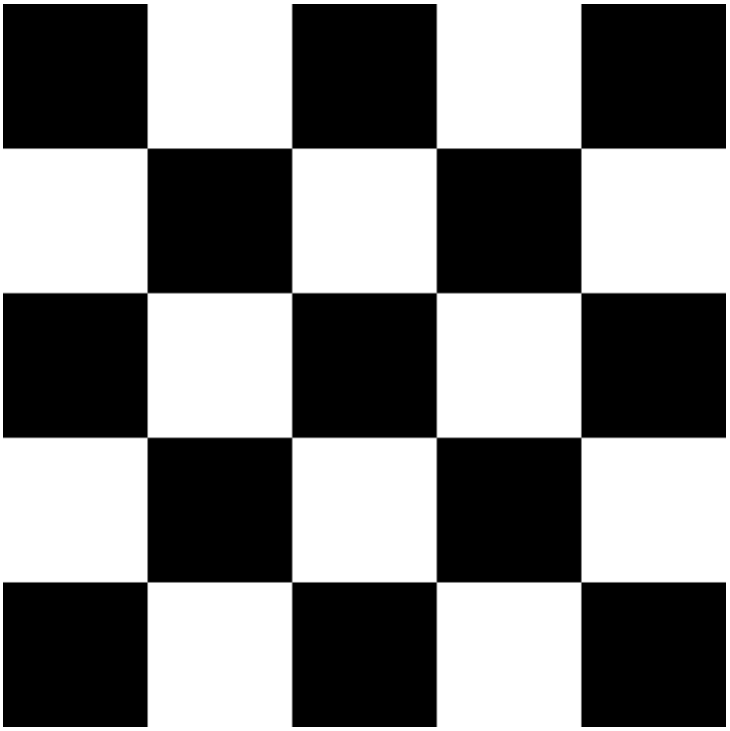
In [2]:
```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Image
%matplotlib inline
```

In [5]:
```
Image(filename="checkerboard_18x18.png")
```

Out[5]:



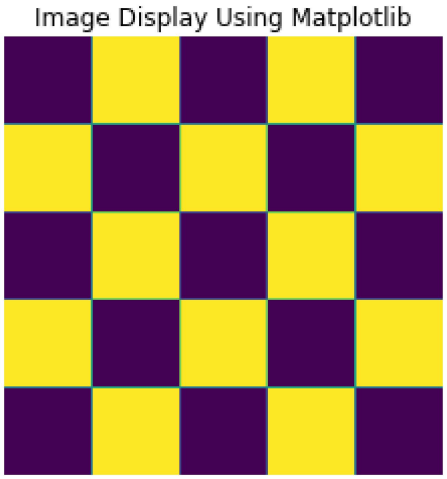
In [6]:
```
# Reading image as gray scale.
cb_img = cv2.imread("checkerboard_18x18.png", cv2.IMREAD_GRAYSCALE)

# Print the image data (pixel values), element of a 2D numpy array.
# Each pixel value is 8-bits [0,255]
print(cb_img)
cv2.imshow('Grayscale Image', cb_img)
```

```
[[0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]
 ...
 [0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]]
```

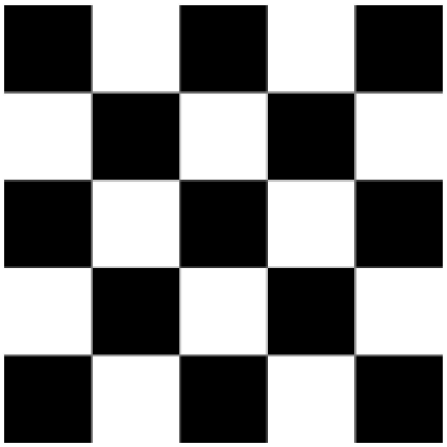
In [7]:
```
# Displaying the image attributes
print("Shape of the image: ", cb_img.shape)
print("Data type of the image: ", cb_img.dtype)
print("Size of the image: ", cb_img.size)
```

```
Shape of the image:  (360, 360)
Data type of the image:  uint8
Size of the image:  129600
```

In [8]:
```
# Displaying image using Matplotlib.
plt.imshow(cb_img)
plt.title("Image Display Using Matplotlib")
plt.axis("off") # To turn off axes
plt.show()
```

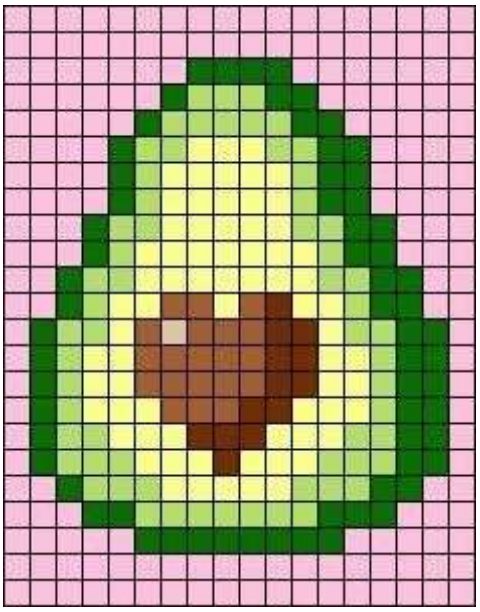


In [9]:
```
# Set color map to gray scale for proper rendering.
plt.imshow(cb_img, cmap = "gray")
plt.axis("off")
plt.show()
```



In [10]:
```
# Read and display image
Image("avacado.jpg")
```

Out[10]:



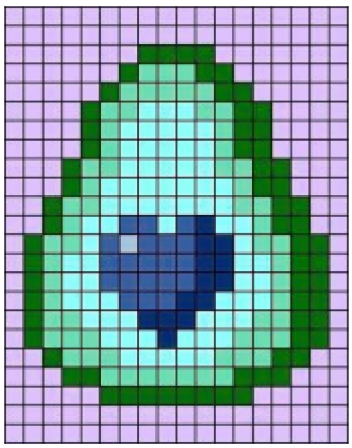
In [11]:
```
# Reading the image
avacado_img = cv2.imread("avacado.jpg", cv2.IMREAD_COLOR)

# Printing the shape of the image
print("Image shape (height, width, channels) is:", avacado_img.shape)
```

```
# Printing the size of the image
print("Image size is: ", avacado_img.size)

# Printing the data-type of the image
print("Data type of image is:", avacado_img.dtype)
```

```
Image shape (height, width, channels) is: (300, 235, 3)
Image size is:  211500
Data type of image is: uint8
```

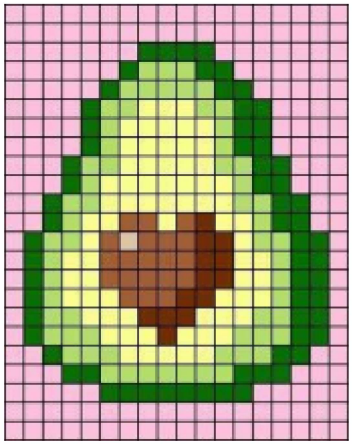
In [12]:

```
# Displaying the image using matplotlib
plt.imshow(avacado_img)
plt.axis("off")
plt.show()
```



In [13]:

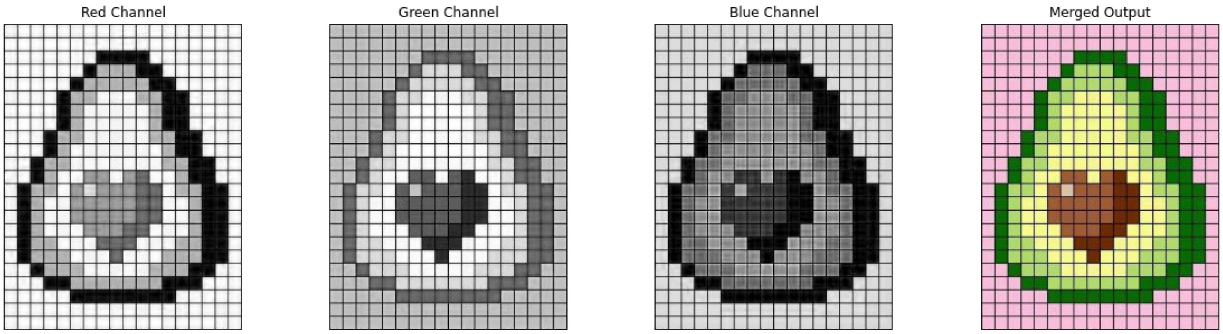
```
# Reversing the channels of the color image
avacado_img_channels_reversed = avacado_img[:, :, ::-1]
plt.imshow(avacado_img_channels_reversed)
plt.axis("off")
plt.show()
```



In [14]:

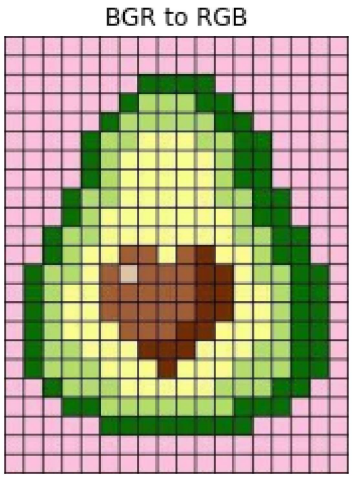
```
# Split the image into the B,G,R components
b, g, r = cv2.split(avacado_img)

# Show the channels
plt.figure(figsize = [20, 5])

# Red Channel
plt.subplot(1, 4, 1)
plt.imshow(r, cmap = "gray")
plt.title("Red Channel")
plt.axis("off")
```

```
# Green Channel
plt.subplot(1, 4, 2)
plt.imshow(g, cmap = "gray")
plt.title("Green Channel")
plt.axis("off")

# Blue Channel
plt.subplot(1, 4, 3)
plt.imshow(b, cmap = "gray")
plt.title("Blue Channel")
plt.axis("off")

# Merge the individual channels into a BGR image
merged_img = cv2.merge((b, g, r))

# Show the merged output
plt.subplot(1, 4, 4)
plt.imshow(merged_img[:, :, ::-1])
plt.title("Merged Output")
plt.axis("off")
plt.show()
```



In [15]:

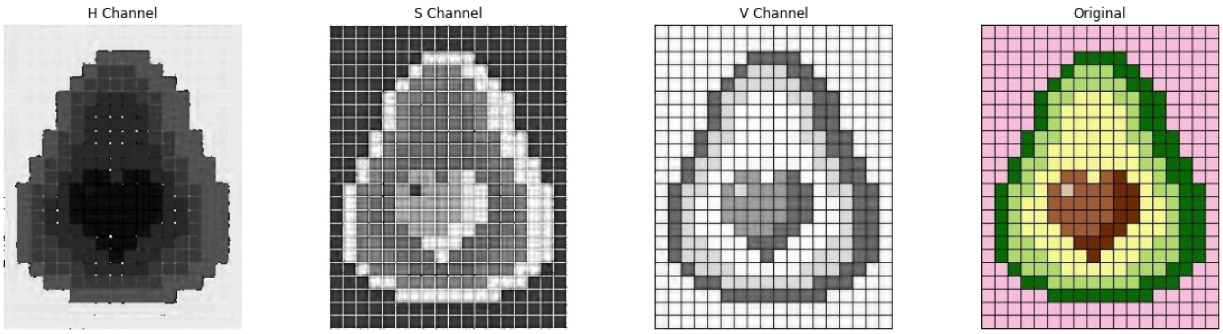
```
# OpenCV stores color channels in a differnet order than most other applications (BG
avacado_img_bgr = cv2.imread("avacado.jpg", cv2.IMREAD_COLOR)

# Converting BGR to RGB
avacado_img_rgb = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2RGB)
plt.imshow(avacado_img_rgb)
plt.title("BGR to RGB")
plt.axis("off")
plt.show()
```



In [16]:

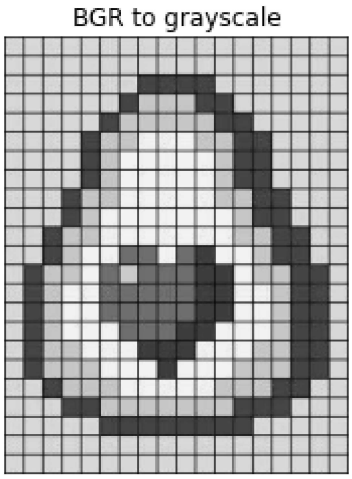
```
# Converting to HSV
avacado_img_hsv = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2HSV)

# Split the image into H, S, V components
h, s, v = cv2.split(avacado_img_hsv)
```

```
# Show the channels
plt.figure(figsize = [20, 5])

# Hue Channel
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H Channel")
plt.axis("off")

# Saturation Channel
plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")

# Value Channel
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
plt.axis("off")

# Show the original image
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Original")
plt.axis("off")
plt.show()
```

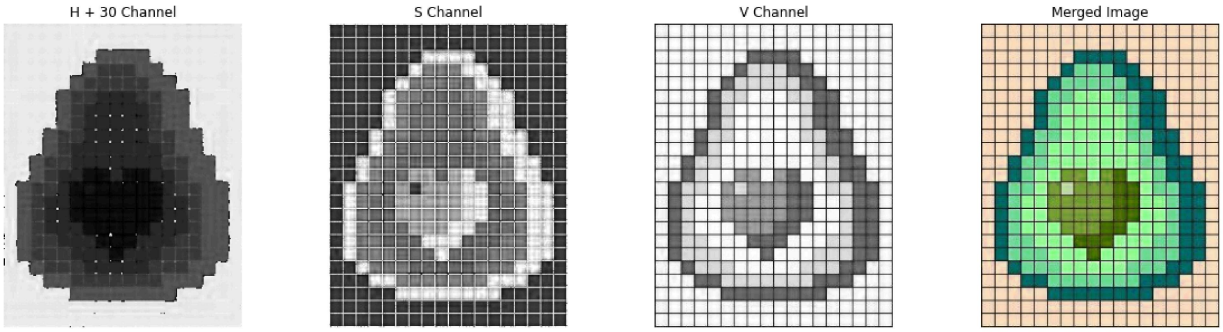


In [18]:

```
# Converting BGR to grayscale
avacado_img_gray = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2GRAY)
plt.imshow(avacado_img_gray, cmap = "gray")
plt.title("BGR to grayscale")
plt.axis("off")
plt.show()
```



In [19]:
```
h_new = h + 30
avacado_img_merged = cv2.merge((h_new, s, v))
avacado_img_rgb = cv2.cvtColor(avacado_img_merged, cv2.COLOR_HSV2RGB)

# Split the image into the B,G,R components
h,s,v = cv2.split(avacado_img_merged)

# Show the channels
plt.figure(figsize = [20, 5])

# Hue Channel
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H + 30 Channel")
plt.axis("off")

# Saturation Channel
plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")

# Value Channel
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
plt.axis("off")

# Show the original image
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Merged Image")
plt.axis("off")
plt.show()
```



In [20]:
```
# Saving the image
cv2.imwrite("avacado_saved.jpg", avacado_img_rgb)
```

Out[20]: True

In [ ]: